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Systems biology-driven analysis of obesity-associated genes with transcript mapping and post-translational modification annotations

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ABSTRACT

Background: Morbid obesity is impacted by several genetic variations, whose functional consequences often go unexamined at the protein level. Despite genome-wide association research identifying numerous loci, post-translational modifications (PTMs) and their interactions remain inadequately comprehended.
Methods: Potential gene candidates and their nonsense single nucleotide polymorphisms (SNPs) were computationally obtained from *dbSNP* database. The mutations were mapped through *SnpEff* with the *GRCh38.86* reference genome to evaluate functional consequences. *Ensembl* genes and transcripts IDs were correlated with *UniProtKB* protein accessions to build protein-variant relationships. PTM characteristics were sourced from *Glygen, SwissPalm, PhosphoSitePlus, ActiveDriverDB, dbPTM* and *UniProt*. An integrated interaction network was built in Cytoscape to connect gene-variants-protein-PTMs-diseases entities. Further, Cytohubba plugin was used to identify high-impact regulatory hubs.
Results: Sixty-six obesity-related genes and their 3581 mutations were retrieved. Functional annotation of these mutations indicated an excess of perturbing missense and frameshift variants in the BBSome, *LEPR*, *GNAS*, and *MC4R* genes. Upon further analysis, a large set of 193 entries from 8 top-ranked proteins was found. *ALMS1, POMC*, and *LEPR* were found to be the main topological hubs in a combined interaction network with 49 nodes and 83 edges. Network analysis showed that phosphoserine and N-glycosylation are important PTM hotspots. Mutational events tend to converge here to impede receptor trafficking and signaling cascades, eventually contributing to hyperphagia and obesity.
Conclusion: Our study links genetic mutations in morbid obesity to protein regulatory pathways which are influenced by PTM alterations. Identifying these topological bottlenecks with a high centrality gives us specific targets for future precision medicine initiatives.

1. Introduction

Morbid obesity is when a person's body mass index (BMI) is 40 kg/m^2 or higher. It is considered to be one of the most severe forms of metabolic disease which is difficult to manage clinically because of biological complexity. The global prevalence of morbid obesity has increased significantly since 1975 with occurrence rates almost tripling in women and rising drastically in men [1]. Predictions suggest that the numbers will rise by more than 100% between 2010 and 2030, which will put significant strain on healthcare systems worldwide, both financially and medically [2,3]. People with morbid obesity are at a higher risk of developing diabetes, cardiovascular diseases, musculoskeletal disorders, and early mortality [4]. Genetic factors (such as rare, high penetrance mutations and common genetic variants) can also affect or increase the susceptibility to morbid obesity. Genome-wide association studies (GWAS) have identified numerous loci associated with BMI and obesity. These include several intronic variants within the FTO gene [5,6]. However, most GWAS findings show statistical associations instead of insights into the biological mechanisms behind genetic variants influencing obesity. GWAS meta-analyses such as the ones conducted by the Genetic Investigation of Anthropometric Traits (GIANT) consortium, have increased the number of obesity-associated loci to more than 536 loci across populations [7,8]. Nevertheless, their functional interpretation remains unexplored. However, studies conducted on the monogenic

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and syndromic forms of obesity have revealed deeper insights. For example, pathogenic mutations (such as the frameshift mutation of BBS9 c.299delC) have linked to Bardet-Biedl syndrome (BBS). These mutations have been identified in affected families and are observed to cause strong functional effects [9,10]. Studies have shown that disruptions in BBS proteins hinders crucial processes like leptin receptor trafficking in neuronal cells. This leads to disturbed energy balance and obesity [11,12]. Nevertheless, functional impacts beyond gene association have not been properly explored for most of the variants linked to obesity.

A major limitation is the lack of integrated workflows that can connect mutations to physiological processes and outcomes. Additionally, many variants discovered by GWAS contribute just a little portion of obesity heredity. They are frequently situated in non-coding regions or within linkage disequilibrium blocks, which complicates the identification of causal variants or processes [13]. Tools like *Variant Effect Predictor (VEP)* [14] and *SnpEff* [15] are commonly used, but they usually focus on predicted transcript-level consequences, not detailed protein level functional effects. Current computational approaches addressing this challenge can be broadly divided into regulatory-based and network-based methods.

Regulatory-based techniques use transcriptomic and epigenomic data to find variants that alter gene expression or transcription factor binding in tissues like the adipose tissue [16]. These methods have been observed to be effective for identifying regulatory variants. However, they fail to provide insight into how changed transcripts can affect protein structure, regulation, or function. Similarly, network-based techniques use interaction data and functional annotations to deduce variant importance. However, they depend on incomplete interaction networks and lack detailed proteomic resolution [17,18]. Only a few studies have investigated obesity-linked variants focusing on alternative transcripts, post-translational modification (PTM) sites, or protein-protein interactions (PPIs). Variants in genes like BBS1 [19], ALMS1 [20], and MC4R [21] have been shown to disturb isoform expression, PTM sites, or important molecular interactions which are involved in energy balance and appetite control [22,23]. However, these studies are generally gene-specific. They also fail to link genetic variants to protein-level features across various obesity-linked genes. Thus, choosing variants for experimental validation remains inefficient and exploratory.

In this work, we present a comprehensive bioinformatics workflow developed to address these gaps. We selected a set of genes linked to morbid obesity and carefully mapped their pathogenic variants using data from *dbSNP* [24] and *ClinVar* [25]. Transcript-level annotations were performed using *SnpEff* [15]. Subsequently, *Ensembl* transcripts [26] were mapped to their *UniProtKB* protein records [27]. Next, we included protein-level information including PTM data from multiple PTM databases, and protein-protein interaction networks from *STRING* database [28]. We developed a unified framework that integrates genetic, transcriptomic, and proteomic information. It was built to understand obesity-associated variants and their functional consequences at the protein level. We found important variant-rich genes (like ALMS1, LEPR), prioritized protein hubs (like POMC, LEPR, and ALMS1), and recurrent PTM hotspots like phosphoserine and N-glycosylation. These all work together on pathways that control hunger, ciliary trafficking, and metabolic signaling. Therefore, we present a data-driven approach for selecting variants for future experimental validation.

2. Methodology

A stepwise computational workflow was used to link morbid obesity-associated genetic variants to potential functional effects at the protein level (Fig. 1). Firstly, a set of morbid obesity-associated genes were curated (n = 66). The gene panel included genes implicated in monogenic obesity, syndromic obesity, and metabolic regulation. These genes were compiled from domain-specific literature and existing curated datasets, as provided in Supplementary Table 1. The chosen gene panel was made to include genes with clear variant-level labels that could be consistently mapped across the dbSNP, transcript, and protein-level databases that were used in this approach. Consequently, some biologically important regulatory genes might not be included because of differences in how detailed the variant annotations were or how consistent the mapping was across different databases. This curated gene list served as the input for variant retrieval in the next step.

2.1. Candidate gene selection and variant retrieval

Next, we aimed to identify genetic variation within the curated gene set. Single-nucleotide variants (SNPs) associated with the selected genes were retrieved from the dbSNP database using automated python queries. Variant extraction was carried out using the Entrez

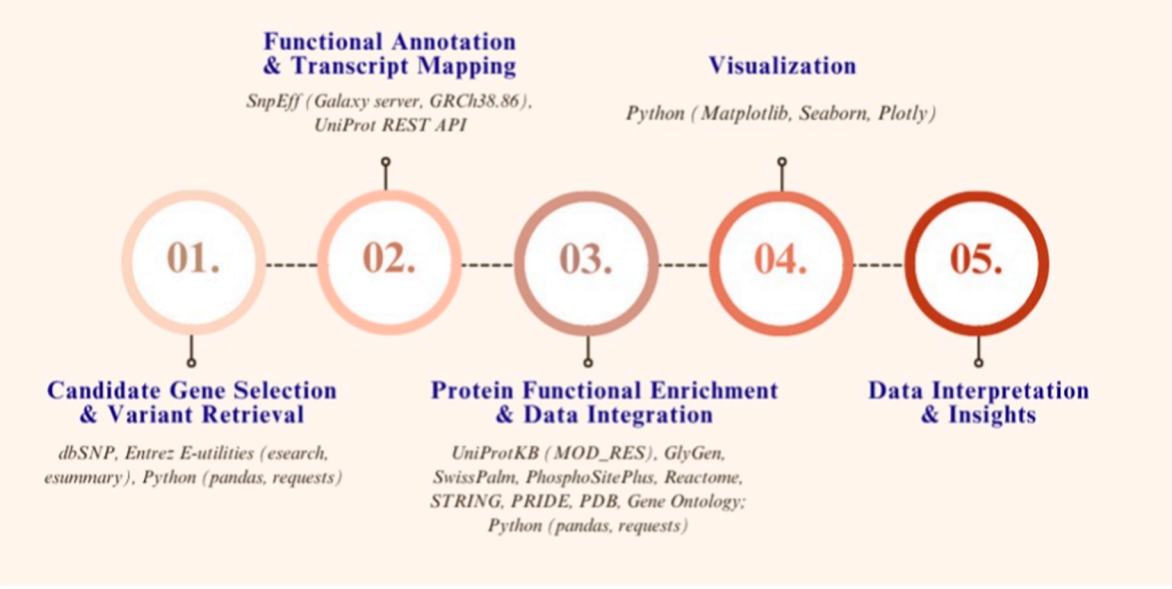


Fig. 1. Workflow and tools utilized in the study. The steps included selection and candidate genes, retrieval of variants, functional annotations, transcript mapping, protein mapping, network analysis, and interpretation.

Programming Utilities (E-utilities) API [29]. Specifically, the *Esearch* function was used to retrieve variant identifiers (rsIDs) for each gene. Queries were executed using the format *"{gene}[gene] AND human[orgn] AND pathogenic [Clinical Significance]*" to restrict retrieval to clinically relevant variants. *ESummary* was then used to download detailed records in XML file format. The XML outputs were parsed to extract key fields like SNP IDs (rsID), chromosomal position, reference alleles, alternate alleles (derived from SPDI notation), functional classification (labelled as FXN_CLASS), clinical notes, and related annotations. This resulted in a total of 3581 unique variants across 66 genes, forming the primary variant dataset. At this point, duplicate rsIDs from different genes were kept to maintain gene-level associations. During downstream integration, deduplication at the variant-level was done wherever required. A delay of 0.3-0.4 s between API calls was implemented to avoid query throttling. This step generated (i) a detailed variant table containing gene-wise SNP annotations; and (ii) a Variant Call Format (VCF) file. The generated VCF file served as the direct input for downstream functional annotation. This file contained essential genomic fields such as CHROM, POS, ID, REF, ALT, and INFO. These variants were then used for downstream annotation. Additionally, gene-specific variant burden was quantified to identify genes with the highest number of associated variants.

2.2. Functional annotation and transcript mapping

The 3581 variants from the VCF file were functionally annotated with *SnpEff v5.2* (release 2023-09-29). These variants resulted in 27,046 transcript-level annotations due to mapping across multiple transcript isoforms and handling of multi-allelic variants. *SnpEff* was executed in a controlled Galaxy server environment [30] using the *GRCh38.86* human reference genome. The analysis was performed using the "*SnpEff -I vcf -o vcf -stats*" command-line configuration with default parameters. *SnpEff* annotated each variant with functional consequence categories and impact classifications. These categories included missense, nonsense, synonymous, frameshift, and splice-site. These annotations included Ensembl transcript IDs (ENST), Ensembl gene IDs (ENSG), and standardized HGVS nomenclature at both coding (HGVS.c) and protein (HGVS.p) levels. Since *SnpEff* outputs are transcript-centric (ENST), these transcripts were mapped to the corresponding UniProt protein entries. This enabled protein-level interpretation. The mapping was performed through the *UniProt* REST API [31] using automated queries. The queries returned important protein metadata like accession numbers, entry names, and sequence attributes.

2.3. Protein functional enrichment and data integration

To enable protein-level interpretation, ENSG and ENST IDs were mapped to *UniProtKB* entries to retrieve protein-level annotations and PTMs. This was performed using *UniProtKB* REST API *idmapping* service [32] in the form of structured ENSG/ENST queries. The general format of the query was:

curl –form 'from = "Ensembl"' \ –form 'to = "UniProtKB‴ \ –form 'ids = "{ENSG or ENST IDs separated by comma}"' \ https://rest.UniProt.org/idmapping/run.

A total of 73 unique ENSG IDs were mapped, which returned 456 UniProtKB results. This formed the core protein dataset. PTM annotations were retrieved from *UniProtKB* using *MOD_RES* feature entries using default parameters, which provide residue-level modification information. To expand PTM coverage, data was further supplemented with information from *Glygen* [33], *SwissPalm* [34], and *PhosphoSitePlus* [35], ActiveDriverDB [36], and dbPTM [37] databases. Gene Ontology (GO) annotations were incorporated to provide functional context for annotated proteins. We used custom Python scripts to integrate results from *SnpEff* annotations with *UniProtKB* and PTM information, along with associated diseases based on PTMs. The final integrated dataset was structured as variant-protein-PTM-diseases associations. PTM annotations from all sources were linked to protein entries based on UniProt accession identifiers. When more than one transcript version mapped to the same protein, variant labels were merged at the protein level to avoid duplication. This produced a non-redundant dataset of variant-associated proteins with corresponding PTM annotations. Appropriate visualizations were created using *matplotlib* [38], *seaborn* [39], and *plotly* [40] python libraries. Overall, this workflow enabled us to systematically map of gene-level variants to functional annotations at the transcript and protein-levels. This led to the creation of a consolidated dataset of reviewed variants, proteins, PTMs, and associated diseases. These represented the functional landscape of genes linked to morbid obesity.

2.4. Multi-layer interaction network construction

Next, a multi-layer network was constructed linking genes, variant, proteins, and PTMs. The integrated dataset from the previous phase included a mapped combination of genes, variant IDs (rsIDs), UniProt entries, and PTM annotations. The dataset was filtered such that it only kept records with reviewed and approved UniProt entries, high-impact variants, proteins with a total PTM count of 10 or more, and a non-zero PTM mutation load. As a result, these mappings were turned directly into edges that showed relationships between genes and variants, variants and proteins, and proteins and PTMs for building networks. Top 30 genes ranked by maximum PTM burden were selected to reduce network complexity. Duplicate gene-variant mappings were removed, and the network was structured hierarchically. Gene-variant associations were derived from dbSNP mappings. Variant-protein associations were obtained from UniProt-based mapping. Protein-PTM and PTM-diseases associations were extracted from UniProt and integrated PTM databases. Additionally, protein-protein interaction data was incorporated from the STRING database by selecting medium to high-confidence interactions (combined score $\geq$0.4). The final network was compiled and imported into Cytoscape for further network visualization. Cytohubba plugin was used to identify top ten hub nodes in the built network based on Betweenness measure of centrality. The betweenness centrality measures a node's importance in a network based on how often a node acts as a bridge along the shortest path between two other nodes.

3. Results

The workflow started with a list of 66 genes associated with morbid obesity. They were carefully selected based on their potential or confirmed involvement in energy balance, adiposity control, and neuroendocrine signaling.

3.1. Variant identification from dbSNP

The list of genes was used as a query input on the *dbSNP* database. The query revealed 3581 unique variants. The results indicated significant genetic variability among obesity-related loci. Further, we classified the variants based on their pathogenicity nature. Classification of pathogenicity was based on dbSNP database's clinical importance remarks. Variants that were marked as "pathogenic" or "likely pathogenic" were put together. All other variants were put into three groups: benign, likely benign, and of unclear significance. Out of all variants, 3028 were identified as pathogenic or possibly pathogenic. This demonstrated a significant presence of functionally harmful mutations in the gene set. The rest of the variants were identified as neutral, likely neutral, or of unclear importance. Further, *dbSNP* identified 1004 missense modifications, 680 frameshift events, and 288 stop-gained variants. The distribution of variants across the gene set was non-uniform. This is because of gene-specific mutational clustering. This happens when some genes get higher reported variants than others because of factors like gene length, functional significance, and the

degree of clinical study. This results in multiple variants being linked to each gene locus. Additionally, certain loci were observed to show high mutational density (Fig. 2), as described below.

3.2. Functional annotation with SnpEff

The variants were annotated by *SnpEff* v5.2 using the *Galaxy* server. This resulted in a total of 27,046 transcript-wide effects. Further, the variants were classified based on their functional impacts. 44.6% of these variants were identified as high-impact variants. These comprised of 12,082 occurrences of frameshift, stop-gain, and splice-site modifications. Similarly, 43.5% of variants were identified as modifier variants in non-coding regions with 11,754 occurrences. 7.9% of variants were identified as moderate-impact missense modifications with 2133 occurrences. Lastly, 4% of the variants were identified as low-impact synonymous modifications with 1077 occurrences. Further, we observed that nonsense variants (53.1%) were more common than missense variants (42.4%). However, only 4.5% of them were silent substitutions, giving missense-to-silent ratio of 9.4. Further, we observed that exonic regions made up 70.1% of the coding regions, followed by splice-site (12.4%) and intronic (10.1%) coding regions. The transition/transversion ratio was observed to be 1.85, as expected from human data. Fig. 3 shows the outcomes of the SnpEff analysis including distribution of variant impacts (Fig. 3a), top ten variant effects (Fig. 3b), variant types (Fig. 3c), and indel length distribution (Fig. 3d) across all genes.

3.3. Transcript-to-protein mapping

In this step, we transitioned from genetic variability to molecular elucidation at the protein level. This enabled us to associate genes with their translational potential. The *Ensembl* to *UniProtKB* accession mapping resulted in 456 gene-to-UniProt and 348 transcript-to-protein links. Subsequent downstream filtering for high-confidence associations resulted in a refined subset of 8 primary entries. These included *Pro-opiomelanocortin* or *POMC* (P01189), *Ectonucleotide pyrophosphatase* or *ENPP1* (P22413), *Malate dehydrogenase* (P40925), *Eukaryotic translation initiation factor 2 subunit 3* (P41091), *Leptin receptor* or *LEPR* (P48357), *BDNF/NT-3 growth factors receptor* or *NTRK2* (Q16620), *Centrosome-associated protein* or *ALMS1* (Q8TCU4), and *Ribosomal biogenesis protein* or *LAS1L* (Q9Y4W2). These proteins were prioritized based on their significant PTM-to-variant ratios. Among these, the LEPR and ALMS1 proteins served as major translational hubs. They connected multiple genetic variants to specific protein isoforms.

3.4. Functional enrichment of mapped proteins

The integration of protein-level ontologies offered comprehensive physiological information for obesity-associated variants. Mapping across transcript isoforms resulted in 27,046 transcript-level annotations. These were consolidated into protein-level mappings. Subsequently, they were filtered to high-confidence protein entries. Functional enrichment analysis of the mapped proteins reviewed a significant clustering in pathways managing energy homeostasis. Impacted proteins were predominantly enriched in pathways such as signal transduction (*LEPR*), lipid metabolism (*PPARG*), and BBSome-mediated ciliary trafficking (*BBS1, BBS2, BBS4*). Quantitative assessment of the final dataset identified a prioritized set of 8 high-confidence proteins. The mutational burden was highest in *ALMS1* with 477 variants. This was followed by *ENPP1* (37 variants), and POMC (12) variants. All these genes (*LEPR, PPARG, BBS1, BBS2, BBS4, ALMS1, ENPP1,* and *POMC*) provided a curated set of 193 entries. This set served as the high-resolution basis for understanding genomic-to-proteomic translation in morbid obesity. However, this approach relies on holistic annotation from curated databases instead of statistical enrichment evaluation. Consequently, pathway correlations are evaluated qualitatively. We focused on biological coherence and frequent occurrence across datasets without the application of statistical adjustments. Additionally, we linked variant-associated proteins to modification descriptors from PTM databases to map residue-level PTMs (e.g., phosphorylation and

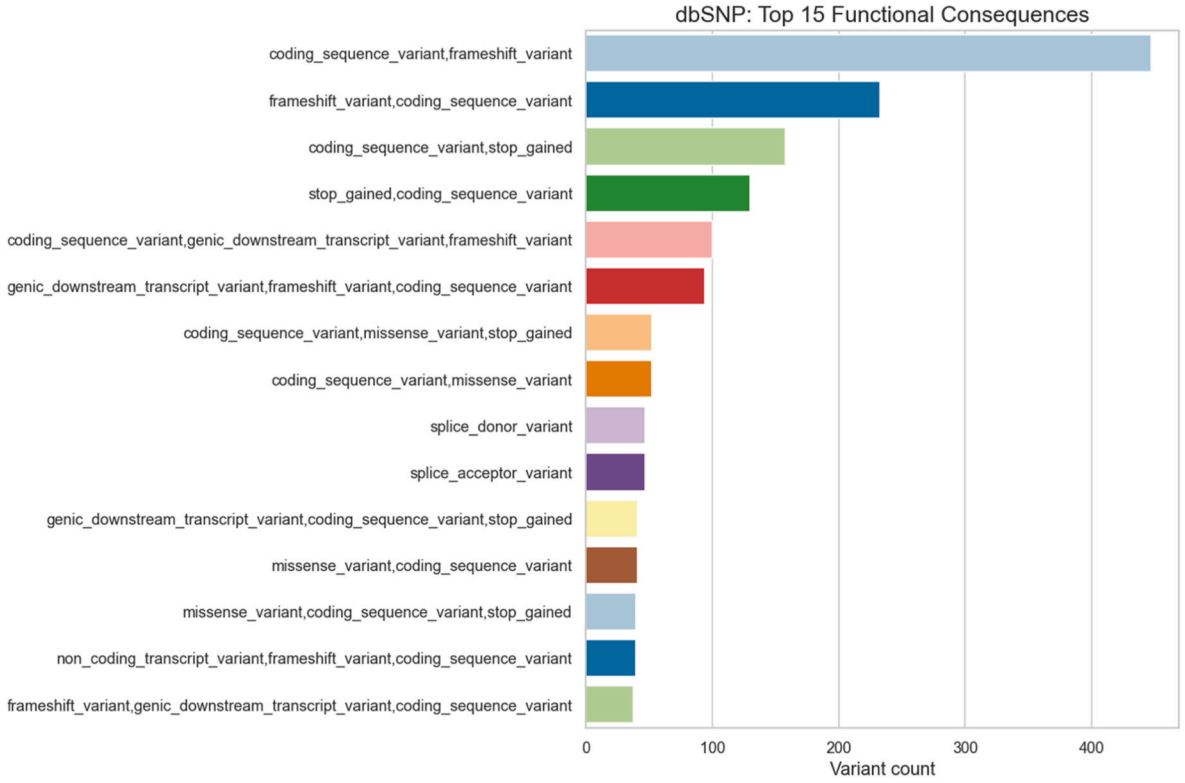


Fig. 2. Top 15 Functional Consequences of identified variants as predicted by the dbSNP analysis.

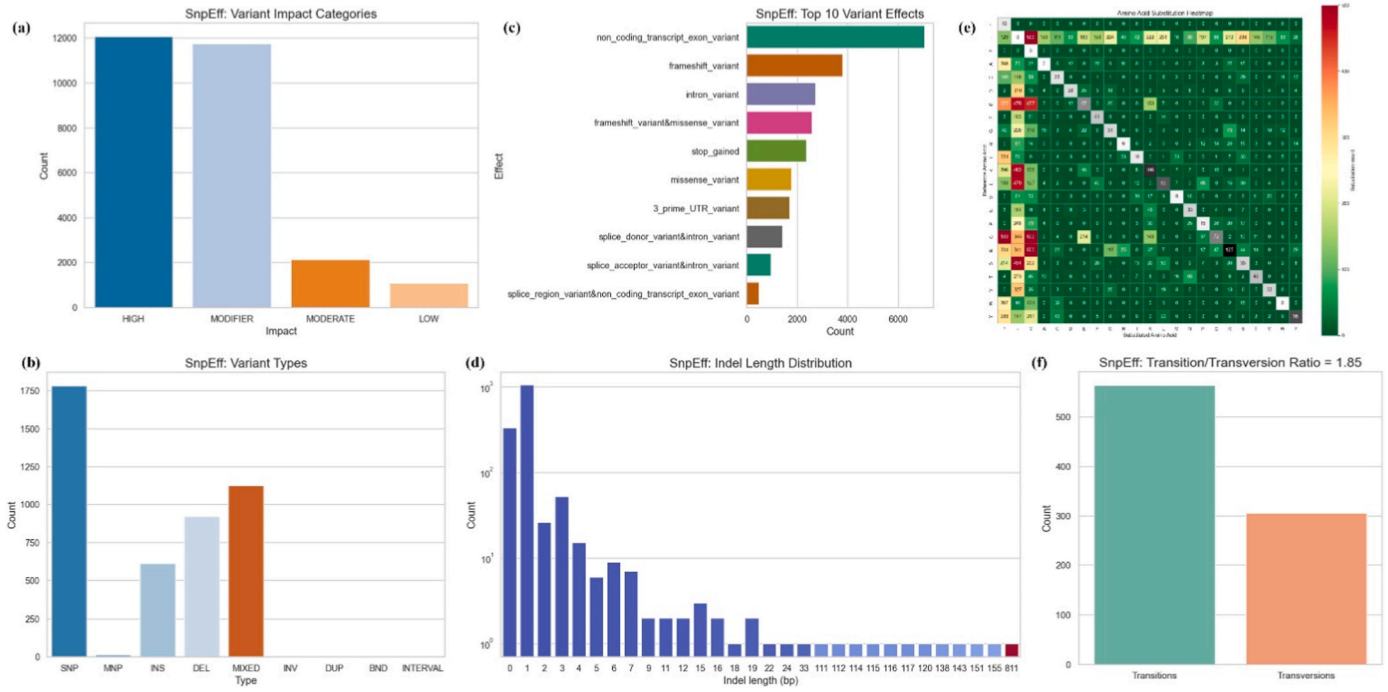


Fig. 3. Variant Annotations using SnpEff revealed distributions of **(a)** Variant Impact Categories; **(b)** Top 10 Variant Effects; **(c)** Variant Types; and **(d)** Indel Length Distribution.

glycosylation sites). Supplementary Table 2 provides detailed variant-protein-PTM associations. It shows genome coordinates, transcript annotations, UniProt protein mappings, PTM types (like phosphorylation and glycosylation), and annotations from PTM databases.

3.5. *Post-translational modification landscape*

Subsequently, PTM patterns were observed, which revealed a non-random enrichment within specific proteins. These proteins included *GNAS* (a *G-protein subunit*) and constituents of the BBSome complex (including *BBS1, BBS4,* and *BBS9*). BBSome proteins had several altered amino acid residues, disulfide linkages, and cross-linked structures. This suggested structural stability and controlled trafficking functions. Further, glycosylation regions were prevalent in BBS1 and BBS9. This aligned with the recognized significance of glycans in ciliary localization and PPIs. Additionally, we observed individual proteins that acted as modification hotspots. The highest PTM-mutation burden was seen in *Centrosome-associated protein ALMS1* (n = 251). This was followed by *Leptin receptor* (n = 92), and *Ribosomal biogenesis protein LAS1L* (n = 72). The PTM-to-variant ratio also showed how functionally dense certain loci were. For example, *Malate dehydrogenase* and *Ribosomal biogenesis protein LAS1L* both had a ratio of 12.0. This showed that even small changes in the patterns of modification along the ciliary and metabolic planes can cause big problems with the energy balance of the whole

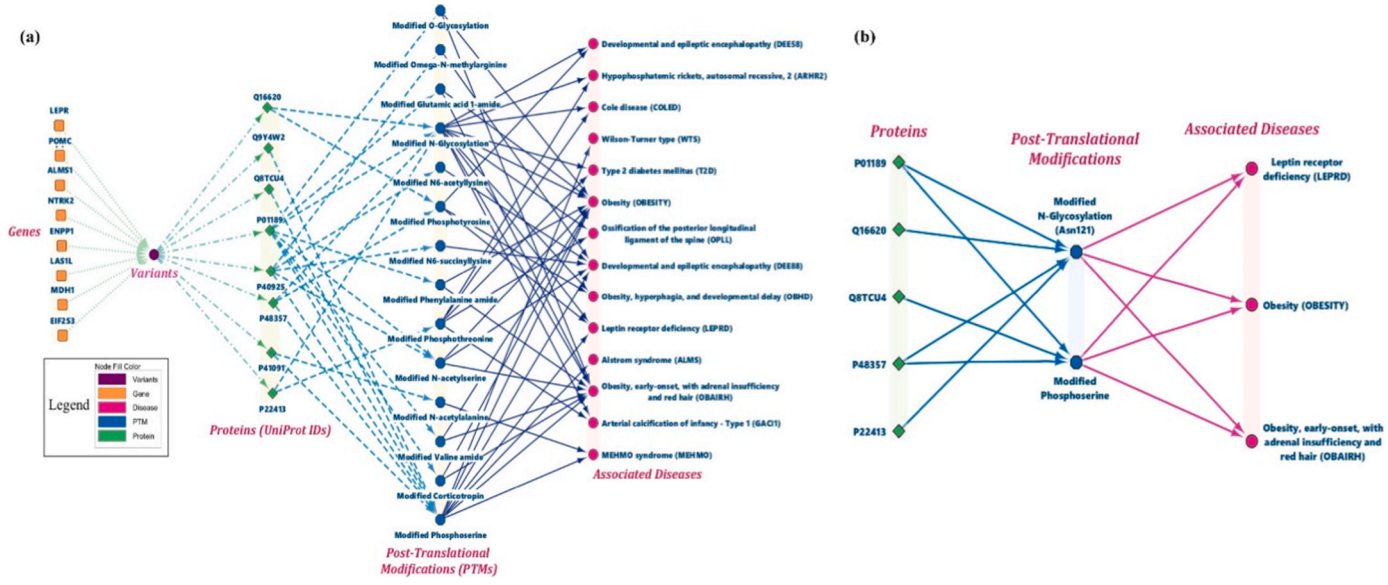


Fig. 4. **(a)** Integrated Interaction Network mapping the associations between prioritized obesity-related genes, genomic variants, UniProt proteins, and PTM sites; **(b)** Identification of top 10 hub nodes based on betweenness centrality with key regulatory proteins, PTMs, and associated diseases.

system. Additionally, PTM-related variation patterns were seen in important receptors like *MC4R* and *LEPR*, which support the regulation of pathways that control hunger and energy equilibrium.

3.6. Integrated interaction network analysis

A combined interaction network was built to show how genetic variation and protein function affect each other in a complex manner. The final network had 49 nodes, 83 edges, and an average of 3.689 neighbors per node (Fig. 4a). The network has a diameter of 5 and characteristic path length of 2.809. This shows that the integrated network structure has a moderate degree of connectivity. We observed two main functional axes. First, a ciliary trafficking axis concentrated on *ALMS1* and *LAS1L*; and second, a neuroendocrine axis dominated by POMC and LEPR. Further, topological evaluation using betweenness centrality identified the top ten primary hub nodes (Fig. 4b). The nodes included 5 proteins, 2 PTMs, and 3 associated diseases. These included *Centrosome-associated protein ALMS1 (*Q8TCU4*), Ectonucleotide pyrophosphatase (*P22413*), Pro-opiomelanocortin (*P01189*), Leptin receptor (*P48357*), and BDNF/NT-3 growth factors receptor (*Q16620*)* proteins, Modified Phosphoserine and Modified N-Glycosylation PTMs, and Obesity, Leptin receptor deficiency (LEPRD), and Obesity, early-onset, with adrenal insufficiency and red hair (OBAIRH) diseases.

4. Discussion

Our work improves the genetic understanding of morbid obesity by methodically merging variant analysis, transcript-protein annotation, and PTM characterization of 66 morbid obesity-associated genes. Previous GWAS and genomics studies have identified disrupting alleles. However, these studies fail to connect mutations to the resulting protein functionalities which are relevant to morbid obesity. In our workflow, we characterize an integrated network of 49 nodes and 83 edges. This enabled the mapping the functional consequences of 193 prioritized variant-protein associations. Our research showed that the development of severe obesity is mainly reliant on a small group of high-centrality hubs. These included *ALMS1, POMC, LEPR,* and *ENPP1*. These hubs serve as junction places for PTM-mediated signaling and structural trafficking. This convergence shows that morbid obesity is more likely to develop due to mutations that change the PTM processes of protein transport, function, and signaling.

Further, we identified significant number of variants in the *MC4R* gene. This gene translates the melanocortin-4 receptor, which is a key controller of appetite [41]. The gene functions as a G-coupled receptor (GPCR) that subdues appetite upon activation by α-MSH in neurons of the hypothalamus [42]. Further, numerous missense variants in *MC4R* were observed to be clustered at anticipated phosphorylation sites. These sites are crucial for receptor deactivation by β-arrestin [43]. The receptor fails to switch off when mutations occur at this site. As a result, hunger signals continue for longer than normal. This leads to persistent appetite stimulation [43]. This disruption further delays the negative feedback mechanism that reduces hunger after food intake [44]. Consequently, the affected individuals experience hyperphagia, which contributes to morbid obesity. Thus, *MC4R* PTM-affecting mutations prevent normal phosphorylation-dependent appetite control, contributing to morbid obesity [21,43].

A similar mechanism was observed for *LEPR*, the gene encoding the leptin receptor [45]. Leptin signaling provides a constant signal to the hypothalamus about body fat stores. It also aids in energy balance within the body. When this signaling pathway is disrupted, food intake becomes uncontrolled, leading to rapid onset of obesity [46,47]. We identified several damaging variants in the *LEPR* gene including p. Trp625*, p.Thr85fs, p.Tyr155fs. These mutations often produce a non-functional or severely shortened leptin receptor. As a result, the brain fails to receive proper satiety signals from the leptin hormone [48, 49]. This further causes *LEPR* deficiency, which is a rare but serious form of monogenic obesity. It is characterized by extreme hyperphagia and rapid weight gain beginning in early infancy [45,50]. Thus, our findings expand the current knowledge of leptin resistance mechanisms by directly linking disease-causing variants to PTM-sensitive residues.

Interestingly, we also detected concentrated phosphorylation hotspots in *GNAS* gene, which encodes the stimulatory *G-protein α-subunit (Gαs)*. *GNAS* mutations are traditionally linked to endocrine disorders such as pseudohypoparathyroidism [51,52]. However, their role in morbid obesity still remains unexplored.

Our PTM analysis showed that obesity-associated *GNAS* mutations were linked to phosphorylation hotspots, that can affect *cAMP* signaling [53]. *cAMP* serves as an additional messenger in leptin [54] and *MC4R* signaling. Thus, disruptions of *GNAS* PTM regulation can reduce appetite-regulating responses. This contributes to morbid obesity due to compromised neuroendocrine signaling [55]. These findings reposition *GNAS* from just a hormone marker to a potential obesity driver. Furthermore, we observed mutations in *BBS* genes (such as *BBS1*, *BBS4*, and *BBS9*). These mutations are causative factors for Bardet-Biedl syndrome, which is a ciliopathy marked by polydactyly, obesity, and degenerative retinal conditions [56]. We identified conserved glycosylation and disulfide bridge patterns in BBSome proteins at the exact locations of disruptive variations. These PTMs are important for preserving structural integrity. They also facilitate the transport of cargo proteins to the cilium, including the leptin receptor [22]. Disruptions can undermine BBSome scaffolding, hinder receptor transport, and exacerbate the typical obesity manifestation [12]. Previous research has shown that ciliary failure is a cause of hyperphagia [57]. However, our findings show that *Centrosome-associated protein ALMS1* and *Ribosomal biogenesis protein LAS1L* are the main contributors in this axis. It is important to note that *ALMS1*, had the most mutations (477 variants) and the most PTM mutations (251). ALMS1 is an important structural protein that is found at the proximal end of centrosomes and basal bodies [58]. It controls the organization of microtubules, the activity of cilia, and the adhesion of centrosomes. Since it is so important to the shape of cells, mutations can cause Alström syndrome, a disease that affects multiple organs [58]. Alström syndrome is characterized by obesity, among other comorbidities [59]. Similarly, mutations in the *LAS1L* gene, located on the X chromosome, can cause Wilson-Turner Syndrome (WTS) [60]. Affected individuals might have severe brain disability, dysmorphic faces, low testosterone levels, short stature, and truncal obesity [60]. The obesity seen in these cases is usually part of a larger syndromic picture rather than dietary-driven obesity [60]. Further, we identified hub nodes with proteins such as *POMC, LEPR, NTRK2, ENPP1,* and PTMs like phosphoserine and N-glycosylation. *POMC-LEPR* signaling disruption affects melanocortin-mediated appetite control, causing hyperphagia and early-onset obesity [61]. The effect is worsened by *NTRK2* dysfunction affecting the *BDNF*-dependent neuronal control of energy balance, and increasing food intake and weight gain [62]. Additionally, *ENPP1*-mediated insulin resistance decreases metabolic efficiency and promotes fat accumulation [63]. Enhanced phosphorylation and glycosylation across these hubs imply that altered PTMs of these pathways increases their combined influence on obesity traits.

Thus, our findings reveal disruptions and alterations in 66 genes converging on proteins such as *LEPR, MC4R, GNAS*, and elements of the BBSome complex, suggesting a polygenic basis for morbid obesity. The results indicate that the mutations are responsible for PTM-associated signaling hubs and interplay between the proteins involved. Previous studies have demonstrated similar results indicating that PTM localization serves as a molecular connection. Our approach is unique not because it looks at individual PTMs or genes, but because it maps out how variants and PTMs are convergent across many obesity-related proteins. It shows that PTM hotspots (like phosphoserine and N-glycosylation) are functional bottlenecks that affect many proteins. Further, GWAS identifies loci typically without figuring out how they work. However, our approach helped us find where variants are located on

functional residues, especially PTM sites. This helped us identify experimentally testable targets. Nonetheless, this research has certain limitations. Not all PTMs discovered in this investigation are directly influenced by genetic variations. Many of them may result due to secondary regulatory, or environment-related processes that are independent of SNPs. Also, combining different databases may cause differences in the depth of annotations. Consequently, the interactions demonstrated here need to be regarded as functional correlations rather than direct causal linkages.

5. Conclusion

With our study, we have integrated a methodological pipeline which indicates that obesity is a multi-factorial disorder, wherein a large number of genetic mutations can lead to disruption of post translational modifications in proteins causing a disruption in the hunger and appetite pathways. As has been indicated by several studies in the past, these proteins act as a switch for our brain and turn on/off the hunger genes and as a result, the brain stops answering properly to leptin, the key hunger hormone. Consequently, our body feels hungry, in return eating more than needed. In our work, we present a model deciphering the molecular mechanisms in case of morbid obesity, that connects molecular abnormalities to leptin resistance, systemic hyperphagia, and hormonal signaling failure, unifying various genetic disorders within the context of PTM. This study provides an integrative framework linking genetic variants to PTM-associated functional disruptions in morbid obesity.

Data availability statement

Not applicable.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used Quillbot in order to improve the readability and understandability of the article. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication. All authors have read and approved the final manuscript.

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CRediT authorship contribution statement

Sudeepti Kulshrestha: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Priyanka Narad:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bbrc.2026.153942.

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